GIP ASSIGNMENT

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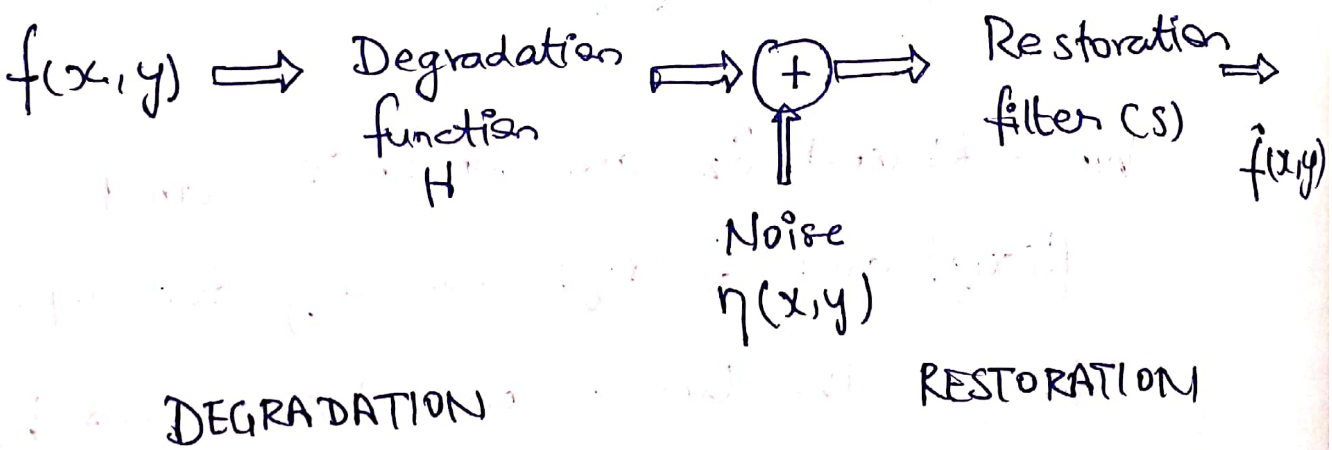
IMAGE RESTORATION

- Image restoration is to restore a degraded image back to the original image.
- Image enhancement is to manipulate the image so that it is suitable for a specific application.

- Identify the degradation process and attempt to reverse it.
- Similar to image enhancement, but more objective.

Model of image degradation



$g(x,y) = f(x,y) * h(x,y) + \eta(x,y)$ – Spatial domain

$G(u,v) = F(u,v) H(u,v) + N(u,v)$ – Frequency domain

$g(x,y) = H[f(x,y) + n(x,y)]$ — (1)

Properties of degradation method

(i) LINEARITY:

$n(x,y) = 0$ in eqn (1)

$\therefore g(x,y) = H[f(x,y)]$ — (2)

H is linear if

$H[K_1 f_1(x,y) + K_2 f_2(x,y)] = K_1 H[f_1(x,y)] + K_2 H[f_2(x,y)]$ — (3)

Where K_1 and K_2 are constants and $f(x,y)$ and $f_2(x,y)$ are any two input images. The linear operator posses both addictive and homogenetity propertity

(ii) ADDICTIVE

If $K_1 = K_2 = 1$ in eq^n (3). It is called addictive property.

$$H[f_1(x,y) + f_2(x,y)] = H[f_1(x,y)] + H[f_2(x,y)] \quad — (4)$$

It states that, if H is a linear operator, the response to a sum of two inputs is equal to the sum of two responses.

(iii) Homogenetity

It states that the response to a constant multiple of any input is equal to the response to that input multiplied by the same constant.

$f_2(x,y) = 0$ in eq^n (3)

$$H[K_1 f_1(x,y)] = K_1 H[f_1(x,y)] \quad — (5)$$

Which is called the homogenity property.

(iv) Position invarient (or) Space invarent

An operation having the input-output relationship $g(x,y) = H[f(x,y)]$ is said to be "position invarient" if

$$H[f(x-\alpha, y-\beta)] = g(x-\alpha, y-\beta) \quad (6)$$

For any $f(x,y)$ and α and β.

<u>Degradation model for Continous function.</u>

When the image $f(x,y)$ is convolved with the two dimensional impulse function $\delta(x,y)$ then it can be represented by

$$f(x,y) = \int_{-\infty}^{\infty} f(x,y)\, \delta(x-x, y-y)\, dx\, dy \quad (7)$$

We use the dummy variable α and β and the above equation can be written as

$$f(x,y) = \int_{-\infty}^{\infty} f(\alpha,\beta)\, \delta(x-\alpha, y-\beta)\, d\alpha\, d\beta \quad (8)$$

in eq ①

$$g(x,y) = H[f(x,y)\} + n(x,y)$$

$$f(x,y) = H\left[\int_{-\infty}^{\infty} f(\alpha,\beta)\, \delta(x-\alpha, y-\beta)\, \partial\alpha\, \partial\beta \right] + n(x,y) \quad -⑨$$

If $n(x,y) = 0$,

$$g(x,y) = \int_{-\infty}^{\infty} H[f(\alpha,\beta)\, \alpha(x-\alpha, y-\beta)]\, \partial\alpha\, d\beta \quad ⑩$$

Since $f(\alpha,\beta)$ is independent of x and y

$$g(x,y) = \int_{-\infty}^{\infty} f(\alpha,\beta)\, H[\partial(x-\alpha, y-\beta)]\, \partial\alpha\, d\beta \quad ⑪$$

We know the "impulse response" H can be as

$$h(x,\alpha,y,\beta) = H[\partial(x-\alpha, y-\beta)] \quad -⑫$$

Then $g(x,y)$ can be represented by

$$g(x,y) = \int_{-\infty}^{\infty} f(\alpha,\beta)\, h(x,\alpha,y,\beta)\, \partial\alpha\, \partial\beta \quad ⑬$$

This equation is called as 'Super position Integral of the first kind'.

If H is a position invarient, then

$$H[\delta(x-\alpha, y-\beta)] = h(x-\alpha, y-\beta) \quad - (14)$$

Then equation becomes.

$$g(x,y) = \int_{-\infty}^{\infty} f(\alpha,\beta)\, h(x-\alpha, y-\beta)\, d\alpha\, d\beta \quad - (15)$$

In the presence of addictive noise, the expression decribing a linear degration model becomes.

$$g(x,y) = \int_{-\infty}^{\infty} f(\alpha,\beta)\, h(x,\alpha,y-\beta)\, d\alpha\, d\beta + n(x,y) \quad - (16)$$

This is the equation for degradation model for continous image function f(x,y).

Discrete degradation image

Digital images are used in most computer based applications and therefore it is needed to formulate discrete degradation model.

Let two function $f(x)$ and $h(x)$ are sampled uniformly to form arrays of dimension A and B respectively.

To avoid overlapping in the individual periods, a common period "$M \geq A+B-1$" is selected and in the individual function zero are appended after the interval A and B in the function $f(x)$ and $h(x)$ and they are represented as $f_e(x)$ and $h_e(x)$

The convolution of $f_e(x)$ and $h_e(x)$ is given by

$$g_e(x) = \sum_{m=0}^{M-1} f_e(m)\, h_e(x-m) \quad \text{—— (17)}$$

for $x = 0, 1, 2 \cdots M-1$.

Both $f_e(x)$ and $h_e(x)$ are assumed to have a period equal to M. $g_e(x)$ also have period.

$$g = Hf \quad \text{—— (18)}$$

$$f = \begin{bmatrix} f_e(0) \\ f_e(1) \\ f_e(M-1) \end{bmatrix} \quad \text{(19)} \quad \& \quad g = \begin{bmatrix} g_e(0) \\ g_e(1) \\ g_e(M-1) \end{bmatrix} \quad \text{(20)}$$

and H is M x M matrix

$$H = \begin{bmatrix} h_e(0) & h_e(-1) & h_e(-2) & h_e(M+1) \\ h_e(1) & h_e(0) & h_e(-1) & h_e(M+2) \\ h_e(2) & h_e(1) & h_e(0) & h_e(M+3) \\ h_e(M-1) & h_e(M-2) & h_e(M-3) & h_e(0) \end{bmatrix} \quad \text{—— (21)}$$

NOISE MODELS

Error or uncertainty of an image arising from such sources as sensor noise, film grain irregularities, and atmospheric light fluctuations.

These all such effects are called as "noise".

Noise Modelling

During image transmission, the noise can be occurs in the digital images. The peformace of image is affected by various factors such as environmental conditions and quality of the sensing elements themselves.

Some important Noise probability density Functions

1) GAUSSIAN NOISE

The Gaussian random variable 'z' is given by $p(z) = \frac{1}{\sqrt{2\pi}\sigma} \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-\bar{z})^2/2\sigma^2}$

Where z = Intensity

$\bar{z}$ = Mean value of z

σ = Its standard deviation.

σ^2 = Variance of z.

ii) RAYLEIGH NOISE

The Rayleigh noise is Given by

$$p(z) = \begin{cases} \frac{2}{b} \frac{2}{b}(z-a)\, e^{-(z-a)^2/b} & \text{for } z \geq a \\ 0 & \text{for } z < a \end{cases}$$

The mean and varience of this density are given by $\bar{z} = a + \sqrt{\frac{\pi b}{4}}\ \sqrt{\frac{\pi b}{4}}$

And $\sigma^2 = \frac{b(4-\pi)}{4}\ \frac{b(4-\pi)}{4}$

3) ERLANG (GAMMA) NOISE:

The Erlang noise is given by

$$p(z) = \begin{cases} a^b z^{b-1} / (b-1)!\, e^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

Where the parameters are such that $a > 0$, b is a positive integer. The mean and variance are given

$\bar{z} = \frac{b}{a}$ $\frac{b}{a}$

and $\sigma^2 = b/a^2$

4) EXPONENTIAL NOISE:

The exponential noise given by,

$$p(z) = \begin{cases} a e^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

Where $a > 0$.

The mean and variance are

$z = 1/a$ $1/a$

and $\sigma^2 = 1/a^2$

5. UNIFORM NOISE:

The uniform noise is Given by

$$P(z) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq z \leq b \\ 0 & \text{otherwise} \end{cases}$$

The mean and variance are

$$\bar{z} = \frac{1}{b-a} \quad \text{and} \quad \sigma^2 = (b-a)^2/12$$

6) IMPULSE (SALT AND PEPPER) NOISE:

The impulse is given by

$$P(z) = \begin{cases} P_a & \text{for } z = a \\ P_b & \text{for } z = b \\ 0 & \text{otherwise.} \end{cases}$$

Periodic Noise:

Periodic noise in an image arises from electrical or electromechanical interference during image acquisition. This noise is the spatially dependent noise, periodic noise can be reduced significantly through frequency domain filtering

$$\sin(2\pi\mu_0 x + 2\pi V_0 x) \Leftrightarrow j\,\tfrac{1}{2}\left[\delta\delta(u + M\mu_0 V + NV_0) - \delta\delta(\mu - M\mu_0 - V - NV_0)\right]$$

Thus if the amplitude of a sine wave in the spatial domain is strong. we would expect the spectrum of the image is a pair of impulses for each sine wave in the image.

Image Restoration Model

An image restoration model is a type of deep learning model designed to enchance or repair digital images that have been degraded or damaged in some way. These models use techniques like denoising, deblurring, super-quality. They are often used in application like image editing, medical imaging, satellite imaging and survelliance.

Some common technique used in image restoration include.

1. DENOISING:

Removing noise from images caused by factors like light conditions or sensor limitations.

Scanned with CamScanner

2. DEBLURRING.

Sharpening blurry images caused by motion blur or out of focus capture.

3. SUPER RESOLUTION

Increasing the resolution of images to generate high quality versions from low-resolution inputs.

4. INPAINTING

Filling in missing (or) damaged parts of an image based on surrounding information.